5.6 Reaction Kinetics (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.6 Reaction Kinetics (A Level Only)
Difficulty	Medium

Time allowed: 70

Score: /53

Percentage: /100

Question 1a

The initial rate of the reaction of chlorine dioxide, ClO_2, and fluorine, F_2, is measured in a series of experiments at a constant temperature.

$$2\,ClO_2 + F_2 \rightarrow 2ClO_2F$$

The results obtained are shown in Table 1.1.

Table 1.1

experiment	$[ClO_2]$ / mol dm^{-3}	$[F_2]$ / mol dm^{-3}	initial rate / mol dm^{-3} s^{-1}
1	0.010	0.060	2.20×10^{-3}
2	0.025	0.060	to be calculated
3	to be calculated	0.040	7.04×10^{-3}

The rate equation is rate = $k[ClO_2][F_2]$.

i)
Explain what is meant by order of reaction with respect to a particular reagent.

[1]

ii)
Use the results of experiment 1 to calculate the rate constant, k, for this reaction. Include the units of k.

k = ... units .... [2]

iii)
Use the data in Table 1.1 to calculate the initial rate in experiment 2.

initial rate = ... mol dm^{-3} s^{-1} [1]

iv)
Use the data in Table 1.1 to calculate $[ClO_2]$ in experiment 3.

$[ClO_2]$ = ... mol dm^{-3} [1]

[5 marks]

Question 1b

i)

Explain what is meant by rate-determining step.

[1]

ii)

The mechanism of the reaction between ClO_2 and F_2 has two steps.

Suggest equations for the two steps of this mechanism.

step 1 ....

step 2 ....

[1]

iii)

State and explain which of the two steps is the rate-determining step.

rate-determining step = ...

[1]

[3 marks]

Question 1c

Describe the effect of temperature change on the rate of a reaction and the rate constant.

[1 mark]

Question 2a

This question is about the reactions of different unknown compounds.

Two compounds, **X** and **Y**, were reacted together.

The initial rate of reaction was measured when compound **X** and compound **Y** were reacted together. The temperature was kept constant and the results of the experiments are shown in **Table 2.1**.

Table 2.1

experiment	[X] / mol dm^{-3}	[Y] / mol dm^{-3}	rate of reaction / mol dm^{-3} s^{-1}
1	0.030	0.040	4.0×10^{-4}
2	0.045	0.040	6.0×10^{-4}
3	0.045	0.060	9.0×10^{-4}
4	0.060	0.120	2.4×10^{-3}

i)
Use the data in **Table 2.1** to deduce the order of reaction with respect to **X**.

[1]

ii)
State the order of the reaction with respect to **Y**.

[1]

iii)
Determine the overall order of the reaction.

[1]

iv)
Write the rate equation for the reaction.

[1]

[4 marks]

Question 2b

Three separate experiments were carried out at 400K with different starting concentrations of A and B. The results are shown in **Table 2.2.**

Table 2.2

experiment	**[A] / mol dm^{-3}**	**[B] / mol dm^{-3}**	**rate of reaction / mol dm^{-3} s^{-1}**
1	0.50	0.30	7.6×10^{-4}
2	0.25	0.30	1.9×10^{-4}
3	0.25	0.60	3.8×10^{-3}

i)
Deduce the order of reaction with respect to each reactant. Explain your reasoning.

order with respect to [A] ....

order with respect to [B] ....

[2]

ii)
State the rate equation for this reaction. Use the rate equation to calculate the rate constant.

Include the units for the rate constant in your answer.

rate =

rate constant, k = ...

units of k = ...

[4]

[6 marks]

Question 2c

The overall rate equation for a reaction is rate = $k[P]^2[Q]$

i)

State what the units would be of the rate constant, k, in this reaction.

[1]

ii)

State the overall order of the reaction above.

[1]

[2 marks]

Question 2d

Chemists measured the rate of a chemical reaction in a series of experiments between compounds C and D at a fixed temperature as shown in **Table 2.3**.

Table 2.3

experiment	**[C] / mol dm^{-3}**	**[D] / mol dm^{-3}**	**rate of reaction / mol dm^{-3} s^{-1}**
1	0.13	0.12	0.32×10^{-3}
2	0.39	0.12	2.88×10^{-3}
3	0.78	0.24	11.52×10^{-3}

Dedude the order of reaction with respect to

- C ....
- D ....

and write the overall rate equation for this reaction.

[2 marks]

Question 3a

The initial rate of reaction for iodine and propanone in acid solution is measured in a series of experiments at constant temperature.

$$CH_3COCH_3 + I_2 \xrightarrow{H^+} CH_3COCH_2I + HI$$

The rate equation was determined experimentally to be as shown.

$$\text{rate} = k\,[CH_3COCH_3][H^+]$$

i)

State the order of reaction with respect to

- CH_3COCH_3 ....
- I_2 ....
- H^+ ....

and state the overall order of the reaction. ....

[2]

ii)

Explain the role of the acid.

[2]

[4 marks]

Question 3b

The rate of this reaction is 5.40×10^{-3} mol dm^{-3} s^{-1} when

- the concentration of CH_3COCH_3 is 1.50×10^{-2} mol dm^{-3}
- the concentration of I_2 is 1.25×10^{-2} mol dm^{-3}
- the concentration of H^+ is 7.75×10^{-1} mol dm^{-3}.

i)

Calculate the rate constant, *k*, for this reaction. State the units of *k*.

k = ...

units = ...

[2]

ii)

Complete the table by placing one tick (✔) in each row to describe the effect of increasing the temperature on the rate constant and on the rate of reaction.

	decreases	no change	increases
rate constant			
rate of reaction			

[1]

[3 marks]

Question 3c

Fig. 3.1 is produced from the results, which shows how the concentration of I_2 changes during the reaction.

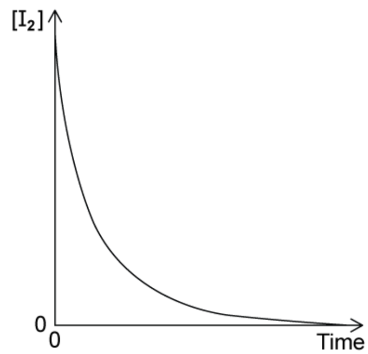


Fig. 3.1

Describe how **Fig 3.1** could be used to determine the initial rate of the reaction.

[2 marks]

Question 3d

On **Fig 3.2**, sketch a graph to show how the initial rate changes with different initial concentrations of H^+ in this reaction.

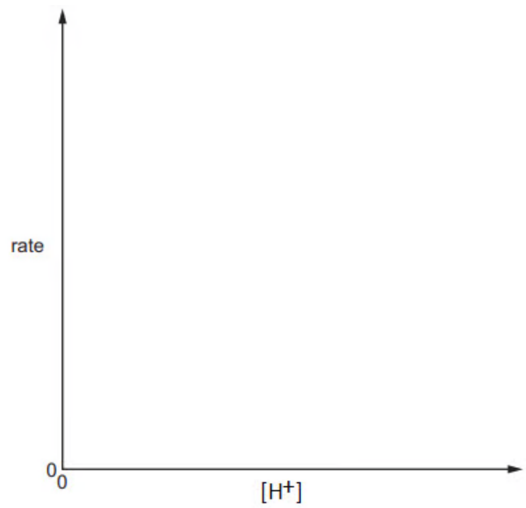


Fig. 3.2

[1 mark]

Question 4a

The equation for the decomposition of hydrogen peroxide without a catalyst is shown.

$$2H_2O_2\,(aq) \rightarrow 2H_2O\,(l) + O_2\,(g)$$

With this information, a student suggests that the rate equation for this reaction is as follows.

$$\text{rate} = k\,[H_2O_2]^2$$

Explain if the student is correct.

[2 marks]

Question 4b

A student carries out separate experiments using different initial concentrations of H_2O_2. The initial rate of each reaction is measured.

The table shows the results that are obtained.

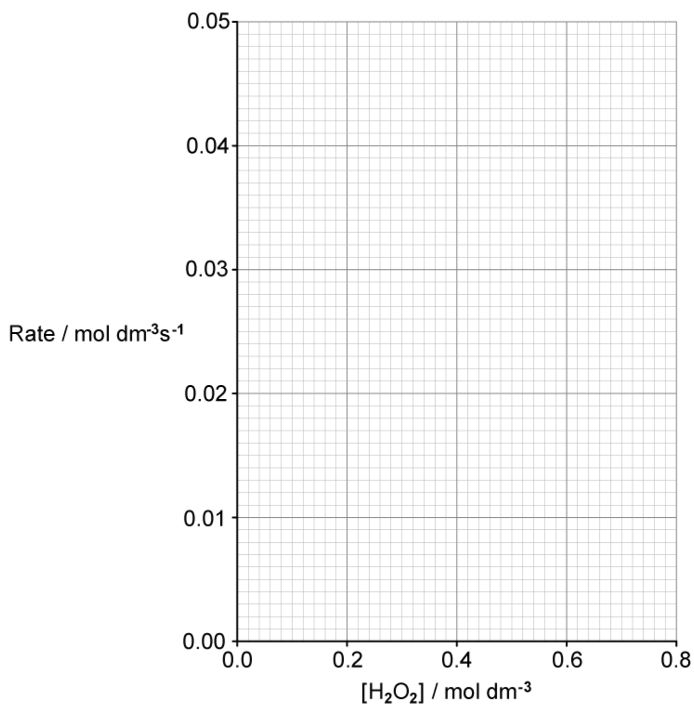
$[H_2O_2]$ / mol dm^{-3}	0.100	0.210	0.285	0.420	0.540	0.700
rate / mol dm^{-3} s^{-1}	0.0055	0.0116	0.0157	0.0230	0.0297	0.0385

i)

Plot a graph on the grid of H_2O_2 concentration against rate of reaction.

Draw a line of best fit through the plotted points.

[2]



ii)

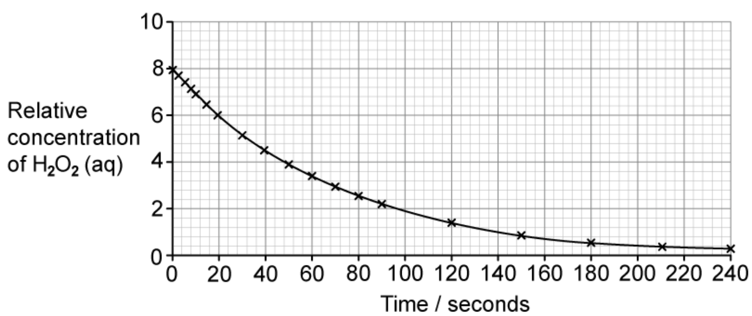
State the rate equation for this reaction.

[1]

[3 marks]

Question 4c

Another student monitors the concentration of H_2O_2 over time. The graph obtained is shown.



Using appropriate calculations, describe how the student could use this graph to prove the order of reaction with respect to H_2O_2.

[2 marks]

Question 4d

Under certain conditions, the decomposition of hydrogen peroxide is found to be first order with respect to hydrogen peroxide, with a rate constant, k, of 2.0×10^{-6} s^{-1}.

i)

Calculate the initial rate of decomposition of a 0.65 mol dm^{-3} hydrogen peroxide solution.

initial rate = ... mol dm^{-3} s^{-1}

[1]

ii)

Explain the effect (if any) of decreasing the temperature on the rate constant and the rate of reaction.

[2]

[3 marks]

Question 5a

This question is about nitrogen oxides.

Nitrogen monoxide is produced by combustion in car engines and released into the atmosphere.

In the lower atmosphere, nitrogen monoxide is responsible for the formation of ozone in a series of reactions shown below.

$$NO(g) + \frac{1}{2}O_2(g) \rightarrow NO_2(g)$$

$$NO_2(g) \rightarrow NO(g) + O(g)$$

$$O_2(g) + O(g) \rightarrow O_3(g)$$

i)
What is the overall equation for this series of reactions?

[1]

ii)
Explain why NO is acting as a catalyst in this reaction.

[1]

[2 marks]

Question 5b

The rate equation for the reaction between nitrogen monoxide (NO) and oxygen (O_2) under certain conditions is given.

$$\text{Rate} = k\,[NO]^2[O_2]$$

The result of an experiment in which NO reacted with O_2 is shown in the table.

initial [NO] / mol dm^{-3}	**initial [O_2] / mol dm^{-3}**	**initial rate of reaction / mol dm^{-3} s^{-1}**
5.0 x 10^{-2}	1.0 x 10^{-2}	6.5 x 10^{-4}

Use the data and the rate equation to calculate a value for the rate constant *k*.

Give the units of *k*.

[2 marks]

Question 5c

Nitrogen dioxide reacts with carbon monoxide at 100 °C to form nitrogen monoxide and carbon dioxide.

$$2NO_2\,(g) + CO\,(g) \rightarrow NO\,(g) + CO_2\,(g)$$

The rate of this reaction was measured at different initial concentrations of the two reagents.

The rate equation was determined to be

$$\text{Rate} = k\,[NO_2]^2$$

The table shows an incomplete set of results.

experiment	$[NO_2]$ / mol dm^{-3}	[CO] / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	4.1×10^{-2}	2.8×10^{-3}	3.3×10^{-5}
2	7.8×10^{-3}	2.8×10^{-3}	to be calculated
3	to be calculated	5.6×10^{-3}	1.8×10^{-4}

i)

Use the data from Experiment 1 to calculate a value for the rate constant, *k*, at this temperature. State its units.

[2]

ii)

Use your value of *k* from (i) to complete the table for the reaction between NO_2 and CO.

[2]

[4 marks]

Question 5d

Nitrogen monoxide, NO (g), reacts with hydrogen, H_2 (g), under certain conditions.

$$2NO\ (g) + 2H_2\ (g) \rightarrow N_2\ (g) + 2H_2O\ (g)$$

The rate equation for this reaction is given.

$$\text{rate} = k\,[NO]^2\,[H_2]$$

One proposed mechanism for this reaction occurs in two steps.

1. $2NO + H_2 \rightarrow N_2 + H_2O_2$
2. $H_2O_2 + H_2 \rightarrow 2H_2O$

Explain which of the steps is the rate-determining step.

[2 marks]

Model Answers: Medium

1a

a)

i) The order of reaction with respect to a particular reagent means:

- The power to which a concentration is raised in the rate equation; [1 mark]

ii) The rate constant, *k*, for this reaction with units is:

- 3.7; [1 mark]
- mol^{-1} dm^{3} s^{-1}; [1 mark]

iii) The initial rate in experiment 2 is:

- 5.50×10^{-3}; [1 mark]

iv) $[ClO_2]$ in experiment 3 is:

- 0.048; [1 mark]

[Total: 5 marks]

- The rate equation for this reaction is rate = $k[ClO_2][F_2]$
- The rate constant is calculated using $k = \frac{\text{rate}}{[ClO_2][F_2]} = \frac{2.20 \times 10^{-3}}{[0.01][0.060]} = 3.7$
- To determine the units:
    - Substitute units into the equation for k
        - $k = \frac{\text{mol dm}^{-3}\text{ s}^{-1}}{(\text{mol dm}^{-3})(\text{mol dm}^{-3})}$
    - Cancel mol dm^{-3}
        - $\frac{\text{s}^{-1}}{(\text{mol dm}^{-3})}$
    - Write the units on one line, but **change the sign of the indices on the bottom:**
        - The units are mol$^{\mathbf{-1}}$ dm$^{\mathbf{3}}$ s^{-1}

- To calculate the initial rate for experiment 2, use the value of k obtained from part i) in the rate equation:
  - rate = 3.7 x [0.02] x [0.06] = 5.50×10^{-3} mol dm^{-3} s^{-1}
- To calculate $[ClO_2]$ in experiment 3, rearrange the rate equation:
  - $[ClO_2] = \frac{7.04 \times 10^{-3}}{3.7\ [0.04]}$ = 0.048 mol dm^{-3}

1b

b)

i) Rate-determining step is:

- The slowest step (in a multi-step reaction); [1 mark]

ii) Equations for the two steps of this mechanism are:

- Step 1: $ClO_2 + F_2 \rightarrow ClO_2F_2$; [1 mark]
- Step 2: $ClO_2 + ClO_2F_2 \rightarrow 2ClO_2F$; [1 mark]

  OR

- Step 1: $ClO_2 + F_2 \rightarrow ClO_2F + F$; [1 mark]
- Step 2: $ClO_2 + F \rightarrow ClO_2F$; [1 mark]

iii) The rate-determining step is:

- Step 1
  AND
  1 mole of F_2 and 1 mole of ClO_2 are reacting / consistent with the rate equation; [1 mark]

[Total: 4 marks]

- The rate-determining step will dictate the overall rate of reaction as it is the slowest step
- To identify the rate-determining step from the equations in the reaction mechanism, look at the rate equation:
  - Rate = $k[ClO_2][F_2]$
  - The species that react in the rate-determining step are the species which are present as concentrations with non-zero orders in the rate equation

        - The order of reaction with respect to ClO_2F_2 and F_2 is 0 so their concentration does not affect the overall rate of reaction
    - The order of reaction with respect to both ClO_2 and F_2 is 1, so their concentration affects the overall rate of reaction and they both appear in Step 1 for each reaction sequence

1c

c) The effect of temperature change on the rate of a reaction and the rate constant is:

- As temperature increases rate increases thus increasing *k*; [1 mark]

[Total: 1 mark]

- The rate constant increases exponentially as temperature increases

2a

a)

i) The order with respect to [X] is;

- 1 / 1st order / $[X]^1$; [1 mark]

ii) The order with respect to [Y] is;

- 1 / 1st order / $[Y]^1$; [1 mark]

iii) The overall order of reaction is;

- (1 + 1 =) 2; [1 mark]

iv) The rate equation for the reaction is;

- rate = k[X][Y]; [1 mark]

[Total: 4 marks]

- When determining the order of a reactant look for a pair of experiments where the concentration of one reactant changes but the concentrations of the other reactants do not change;

- Comparing experiments 1 to 2 for X:
  - The concentration or [X] increases by a factor of 1.5
  - The [Y] remains the same
  - The initial rate increases by a factor of 1.5 - the same as the increase in [X] - meaning the rate is proportional to the [X]
  - So, it is 1st order with respect to [X]
- Comparing experiments 2 and 3 for Y:
  - The [Y] increases by a factor of 1.5
  - The [X] remains the same
  - The initial rate increases by a factor of 1.5 - the same as the increase in [Y] - meaning the rate is proportional to the [Y]
  - So, it is also 1st order with respect to [Y]
- For the rate equation, you must include *k* and [ ] to represent the concentration
  - You don't include first power orders so we write [X] not $[X]^1$

2b

b)

i) The order with respect to each reactant is:

- A = 2 /2nd **/** $[A]^2$; [1 mark]
- B = 1 /1st / [B] / $[B]^1$; [1 mark]

ii) The rate equation for this reaction is:

- rate = $k[A]^2[B]$; [1 mark]

To calculate *k*:

- $k = \frac{\text{rate}}{[A]^2[B]}$
  OR
  $k = \frac{7.6 \times 10^{-4}}{[0.5]^2[0.3]}$; [1 mark]
- *k* = 0.010; [1 mark]
- Units = $mol^{-2}\ dm^{6}\ s^{-1}$; [1 mark]

[Total: 6 marks]

- Firstly you need to determine the rates for [A] and [B]
- If you compare experiments 1 and 2 - the [B] remains the same and the [A] changes
  - The [A] changes from 0.50 to 0.25 which is a factor of 2
  - The initial rate changes from 7.6 x 10^{-4} to 1.9 x 10^{-4} which is a factor of 4
  - So it must be **2nd order with respect to A as when you double the [A] you quadruple the rate**
- If you compare experiments 2 and 3 - the [A] remains the same and [B] changes
  - The [B] changes from 0.30 to 0.60 which is a factor of 2
  - The initial rate changes from 1.9 x 10^{-4} to 3.8 x 10^{-3} which is a factor of 2
  - So it must be **1st order with respect to B as when you double the [B] you double the rate**
- From this, you can then work out the overall rate equation and rearrange it to make k the subject
- It's then a case of choosing an experimental set of data and inserting the values into your calculation for *k*
  - Remember that $[A]^2$ so you need to factor this into your calculation for *k*
- You can use the rate equation to determine the units of *k*:
  - Start with the rearranged rate equation:
    - $k = \frac{rate}{[A]^2\,[B]}$
  - Insert the units:
    - $k = \frac{\text{mol dm}^{-3}\text{s}^{-1}}{[\text{mol dm}^{-3}]^2\,[\text{mol dm}^{-3}]}$
  - One mol dm^{-3} from the bottom cancels the mol dm^{-3} on the top
    - $k = \frac{\text{s}^{-1}}{[\text{mol dm}^{-3}]^2}$
  - These units are then written as mol^{-2} dm^6 s^{-1}

2c

c)

i) The units of the rate constant, *k*, in this reaction are:

- mol^{-2} dm^6 s^{-1}; [1 mark]

ii) The overall order of the reaction is:

- 3; [1 mark]

your calculation for *k*

- Remember that $[A]^2$ so you need to factor this into your calculation for *k*

- You can use the rate equation to determine the units of *k*:
  - Start with the rearranged rate equation:
    - $k = \dfrac{rate}{[A]^2\,[B]}$
  - Insert the units:
    - $k = \dfrac{\text{mol dm}^{-3}\text{s}^{-1}}{[\text{mol dm}^{-3}]^2\,[\text{mol dm}^{-3}]}$
  - One mol dm^{-3} from the bottom cancels the mol dm^{-3} on the top
    - $k = \dfrac{\text{s}^{-1}}{[\text{mol dm}^{-3}]^2}$
  - These units are then written as mol^{-2} dm^{6} s^{-1}

2c

c)

i) The units of the rate constant, *k*, in this reaction are:

- mol^{-2} dm^{6} s^{-1}; [1 mark]

ii) The overall order of the reaction is:

- 3; [1 mark]

[Total: 2 marks]

- To calculate k, you need to rearrange the rate equation;
  - $k = \dfrac{rate}{[P]^2[Q]}$
- You then insert the units for rate (mol dm^{-3} s^{-1}) and units for concentration (mol dm^{-3})
  - $k = \dfrac{\text{mol dm}^{-3}\text{ s}^{-1}}{[\text{mol dm}^{-3}][\text{mol dm}^{-3}][\text{mol dm}^{-3}]}$
- Then cancel mol dm^{-3} where possible
  - $k = \dfrac{\text{s}^{-1}}{[\text{mol dm}^{-3}][\text{mol dm}^{-3}]}$
  - Which can then be written as mol^{-2} dm^{6} s^{-1}
- For calculating the overall order
  - Add all of the indices / powers together for each reactant

  - There is a reactant which is second order and one reactant which is first order
  - So, 2 + 1 = 3

2d

d) The order of the reaction is:

- C = 2 / 2nd **/** $[C]^2$
  AND
  D = 0 / zero / $[D]^0$; [1 mark]

The overall rate equation for this reaction is:

- Rate = $k[C]^2$; [1 mark]

[Total: 2 marks]

- If you compare experiments 1 and 2 - the [D] remains the same and the [C] changes
  - The [C] changes from 0.13 to 0.39 which is a factor of 3
  - The initial rate changes from 0.32×10^{-3} to 2.88×10^{-3} which is a factor of 9
  - C must be **2nd order as you triple the [C] you ninth the rate, or 32**
- For working out the order of D you will notice that there are no experiments where the [C] remains the same as [D] changes - this is a more common exam question
- If you compare experiments 2 and 3;
  - We are unsure whether the initial rate is changing due to the change in [C] or [D] as they are both changing from experiments 2 and 3
  - If you multiplied the initial rate of $2.88 \times 10^{-3} \times 2$ which is the factor by which the [D] is changing, this would give you 5.76×10^{-3} yet the initial rate is 11.52×10^{-3}
  - We know that [C] is 2nd order and when you divide 0.78 by 0.39 this is a factor of 2 and yet the rate changes from 2.88×10^{-3} to 11.52×10^{-3} which is a factor of 4, or 22
  - D must be 0 **order as when you double the [D], the rate changes by 0**
- **Careful:** When you write your final answer, you should not include [D] or $[D]^0$

3a

a)

i) The orders of reaction are :

- CH_3COCH_3 = 1 / 1st order / $[CH_3COCH_3]$ / $[CH_3COCH_3]^1$
  AND
  I_2 = 0 / zero / $[I_2]^0$
  AND
  H^+ = 1 / 1st order / $[H^+]$ / $[H^+]^1$; [1 mark]

The overall order of the reaction is:

- 2 / 2nd; [1 mark]

ii) The role of the acid is:

- A catalyst; [1 mark]
- (Because,) it features in the rate equation but is not a reactant; [1 mark]

[Total: 4 marks]

- For the orders of reaction:
  - CH_3COCH_3 features in the rate equation but is not raised to a power
    - Therefore, it is order 1
  - I_2 does not feature in the rate equation
    - Therefore, it is order 0
  - H^+ features in the rate equation but is not raised to a power
    - Therefore, it is order 1
  - The overall order of the reaction is 1 + 0 + 1 = 2
- You should be aware that any chemical that is not a reactant but does feature in the rate equation is likely to be a catalyst

3b

b)

i) To calculate the rate constant, *k*, for this reaction:

- *k* = 0.46(452); [1 mark]
- Units = $mol^{-1}\ dm^3\ s^{-1}$; [1 mark]

ii) The completed table describing the effect of increasing the temperature on the rate constant and on the rate of reaction is:

	decreases	no change	increases
rate constant			✔
rate of reaction			✔

- Two ticks in the increases boxes as shown; [1 mark]

[Total: 3 marks]

- You were given the rate equation in part (a):
  - rate = $k\,[CH_3COCH_3][H^+]$
- This rearranges to:
  - $k = \dfrac{rate}{[CH_3COCH_3][H^+]}$
- Substituting the given values leads to:
  - $k = \dfrac{5.40 \times 10^{-3}\ \text{mol dm}^{-3}\ \text{s}^{-1}}{[1.50 \times 10^{-2}\ \text{mol dm}^{-3}][7.75 \times 10^{-1}\ \text{mol dm}^{-3}]}$
- This can be evaluated to give k = 0.46 (to two significant figures) with units of $\text{mol}^{-1}\ \text{dm}^{3}\ \text{s}^{-1}$
- When the temperature is increased, a greater proportion of molecules have energy equal to or greater than the activation energy
  - The rate constant and rate of reaction are directly proportional to the fraction of molecules with energy equal to or greater than the activation energy
  - Therefore, at higher temperatures the rate constant and the rate of reaction increase

3c

c) **Fig 3.1** could be used to determine the initial rate of the reaction by:

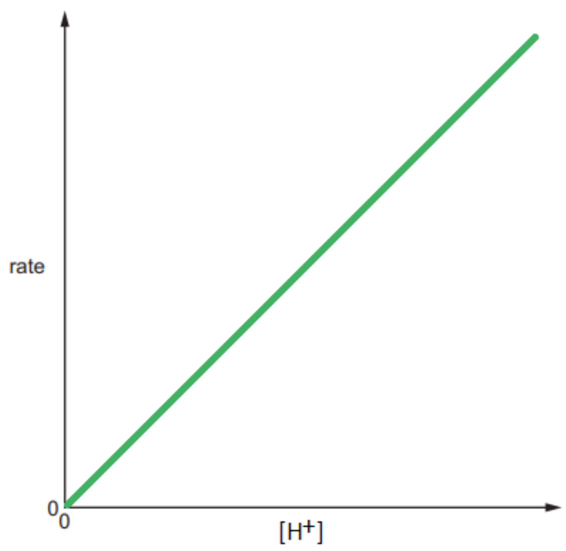
- Drawing a tangent at time, t=0; [1 mark]
- Measuring / calculating the gradient / slope of the tangent; [1 mark]

[Total: 2 marks]

- The initial rate of reaction is at / just after time = 0
- Therefore, to determine the initial rate of reaction you have to draw a tangent to the curve and calculate the gradient

3d

d) The sketch graph showing how the initial rate changes with different initial concentrations of H^+ in this reaction is:



- A straight line starting at the origin / 0,0
  AND
  Showing that rate is proportional to $[H^+]$; [1 mark]

[Total: 1 mark]

- You know from the rate equation, and your answer, in part (a) that the order with respect to H^+ is 1
- Therefore, you need to draw a first order reaction-concentration graph
  - This is shown by a straight line starting at the origin and increasing as you move along the x-axis

4a

a) Explanation of the student's suggestion:

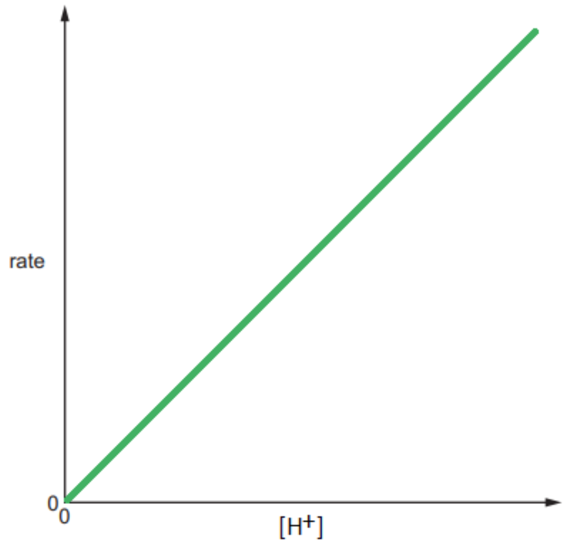
The student is **not** correct because:

- The rate equation cannot be determined from the chemical equation / stoichiometry; [1 mark]
- The rate equation is determined experimentally; [1 mark]

[Total: 2 marks]

- If the answer states that the student is correct then no marks can be awarded

d) The sketch graph showing how the initial rate changes with different initial concentrations of H^+ in this reaction is:



- A straight line starting at the origin / 0,0
  AND
  Showing that rate is proportional to $[H^+]$; [1 mark]

[Total: 1 mark]

- You know from the rate equation, and your answer, in part (a) that the order with respect to H^+ is 1
- Therefore, you need to draw a first order reaction-concentration graph
  - This is shown by a straight line starting at the origin and increasing as you move along the x-axis

4a

a) Explanation of the student's suggestion:

The student is **not** correct because:

- The rate equation cannot be determined from the chemical equation / stoichiometry; [1 mark]
- The rate equation is determined experimentally; [1 mark]

[Total: 2 marks]

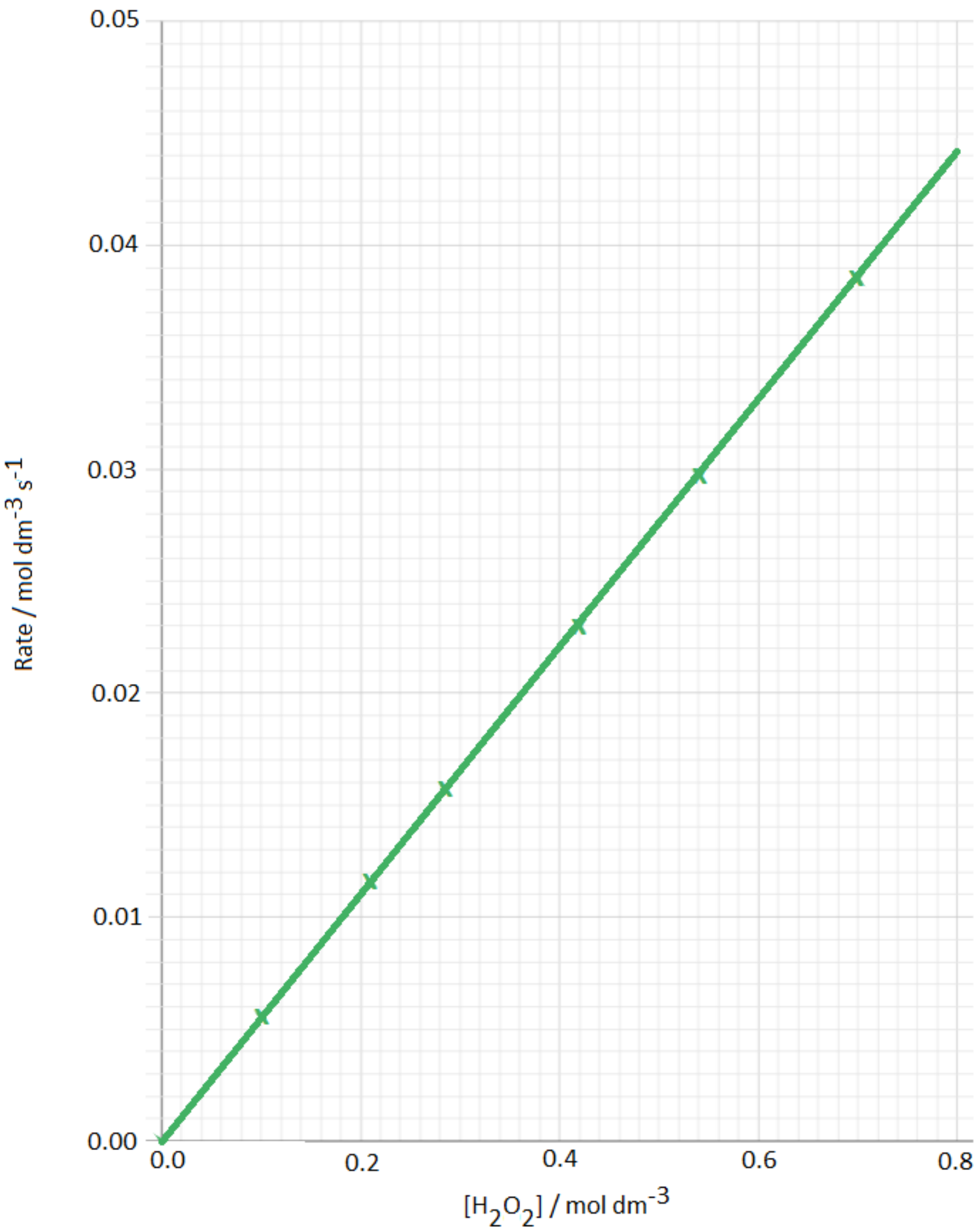
- If the answer states that the student is correct then no marks can be awarded

- This question could potentially be for one mark by the examiner expecting you to give both marking points
- **Remember:** Rate equations can only be determined from experimental results, they cannot be deduced from the chemical equation
    - There are times when the rate equation matches the stoichiometry of the chemical equation but this is coincidental

4b

b)

i) The completed graph is:



- All 6 points plotted correctly; [1 mark]
- A straight line of best fit starting at the origin; [1 mark]

ii) The rate equation for this reaction is:

- Rate = $k[H_2O_2]$; [1 mark]

[Total: 3 marks]

- Although graph plotting questions for reaction kinetics are not common, they are still possible
- For these questions, you must be able to plot the points given accurately
- This question could be for more marks by giving you a completely blank grid and expecting you to scale and label the axes, including units
- For part (ii), the graph is a straight line which indicates that the reaction is first order with respect to H_2O_2
  - Therefore, the rate equation is rate = $k[H_2O_2]$
  - A common mistake for the rate equation would be missing the k

4c

c) How the student could use this graph to prove the order of reaction with respect to H_2O_2:

- Measuring two (or more) half-lives; [1 mark]
- Evidence of two (or more) half-lives calculated / shown on the graph; [1 mark]

[Total: 2 marks]

- The order with respect to H_2O_2 can be proven using half-lives
  - For a zero order reaction, the half-life decreases with decreasing concentration
  - For a first order reaction, the half-life is constant
  - For a second order reaction, the half-life increases with decreasing concentration
- From the previous question, you should be suspecting that the the reaction is first order with respect to H_2O_2
- This would be shown by two (or more) half-lives being the same value
- To achieve both marks, you need to show calculations proving that the half-lives remain constant

- e.g. relative concentration of 8 → relative concentration of 4 goes from 0 seconds to around 47 seconds
  AND
  Relative concentration of 4 → relative concentration of 2 goes from around 47 seconds to around 94 seconds
- Both half-lives are around 47 seconds which means that the reaction can be proven to be first order with respect to H_2O_2

4d

d)

i) The initial rate of decomposition of a 0.65 mol dm^{-3} hydrogen peroxide solution is:

- 1.3 x 10^{-6} (mol dm^{-3} s^{-1}); [1 mark]

ii) How decreasing the temperature will affect the rate constant and the rate of reaction:

- The rate constant and the rate of reaction will decrease; [1 mark]
- (Because,) a smaller proportion of / less molecules will have energy equal to or greater than the activation energy; [1 mark]

[Total: 3 marks]

- The question clearly stated that this specific reaction is first order with respect to H_2O_2
  - Therefore, the rate equation is rate = k [H_2O_2]
- This means that the calculation for the initial rate is rate = 2.0 x 10^{-6} x 0.65 = 1.3 x 10^{-6} mol dm^{-3} s^{-1}
- Since this reaction is first order, decreasing the temperature leads to a smaller proportion of molecules having energy equal to or greater than the activation energy
  - The rate constant and rate of reaction are directly proportional to the fraction of molecules with energy equal to or greater than the activation energy
  - Therefore, at lower temperatures the rate constant and the rate of reaction decrease

5a

a)

i) The overall equation for this series of reactions is:

- $1½O_2 \rightarrow O_3$
  OR
  $3O_2 \rightarrow 2O_3$; [1 mark]

ii) NO is acting as the catalyst because:

- It is used in the first step
  AND
  Regenerated in the second step; [1 mark]

[Total: 2 marks]

- To find the overall equation put all of the reactants together and all of the products together:
  - $NO (g) + ½O_2 (g) + NO_2 (g) + O_2 (g) + O (g) \rightarrow NO_2 (g) + NO (g) + O (g) + O_3 (g)$
- Cancel out any species that appear on both sides of the equation and simplify:
  - $\cancel{NO (g)} + ½O_2 (g) + \cancel{NO_2 (g)} + O_2 (g) + \cancel{O (g)} \rightarrow \cancel{NO_2 (g)} + \cancel{NO (g)} + \cancel{O (g)} + O_3 (g)$
  - $½O_2 (g) + O_2 (g) \rightarrow O_3 (g)$
  - $1½O_2 (g) \rightarrow O_3 (g)$
- One description of a catalyst is a chemical that can be used in a reaction but is not used up
  - This means that a catalyst may be involved in the reaction but it is regenerated at the end of the reaction
    - NO is used in the first reaction equation
    - NO is formed in the second reaction equation

5b

i) The rate constant *k* and its units are :

- *k* = 26; [1 mark]
- $mol^{-2}\ dm^6\ s^{-1}$; [1 mark]

[Total: 2 marks]

- To calculate the rate constant *k*, you have to rearrange the rate equation:
  - Rate = $k\,[NO]^2[O_2] \rightarrow k = \dfrac{\text{rate}}{[NO]^2\,[O_2]}$

i) The overall equation for this series of reactions is:

- $1\frac{1}{2}O_2 \rightarrow O_3$
  OR
  $3O_2 \rightarrow 2O_3$; [1 mark]

ii) NO is acting as the catalyst because:

- It is used in the first step
  AND
  Regenerated in the second step; [1 mark]

[Total: 2 marks]

- To find the overall equation put all of the reactants together and all of the products together:
  - $NO\,(g) + \frac{1}{2}O_2\,(g) + NO_2\,(g) + O_2\,(g) + O\,(g) \rightarrow NO_2\,(g) + NO\,(g) + O\,(g) + O_3\,(g)$
- Cancel out any species that appear on both sides of the equation and simplify:
  - $\cancel{NO\,(g)} + \frac{1}{2}O_2\,(g) + \cancel{NO_2\,(g)} + O_2\,(g) + \cancel{O\,(g)} \rightarrow \cancel{NO_2\,(g)} + \cancel{NO\,(g)} + \cancel{O\,(g)} + O_3\,(g)$
  - $\frac{1}{2}O_2\,(g) + O_2\,(g) \rightarrow O_3\,(g)$
  - $1\frac{1}{2}O_2\,(g) \rightarrow O_3\,(g)$
- One description of a catalyst is a chemical that can be used in a reaction but is not used up
  - This means that a catalyst may be involved in the reaction but it is regenerated at the end of the reaction
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5b

i) The rate constant *k* and its units are :

- k = 26; [1 mark]
- $mol^{-2}\,dm^{6}\,s^{-1}$; [1 mark]

[Total: 2 marks]

- To calculate the rate constant *k*, you have to rearrange the rate equation:
  - $\text{Rate} = k\,[NO]^2[O_2] \rightarrow k = \dfrac{\text{rate}}{[NO]^2\,[O_2]}$

- Inserting the values and evaluating the expression gives:
    - $k = \dfrac{6.5 \times 10^{-4}}{[5 \times 10^{-2}]^2\,[1.0 \times 10^{-2}]} = 26$
- The units can be determined by inserting the units into the equation and cancelling where possible:
    - $k = \dfrac{\text{mol dm}^{-3}\,\text{s}^{-1}}{[\text{mol dm}^{-3}]^2\,[\text{mol dm}^{-3}]} = \dfrac{\text{s}^{-1}}{[\text{mol dm}^{-3}]^2} = \text{mol}^{-2}\,\text{dm}^{6}\,\text{s}^{-1}$

5c

c)

i) The calculated value for the rate constant *k* and its units are:

- *k* = 1.96 x 10^{-2}
  OR
  k = 2.0 x 10^{-2}; [1 mark]
- Units = mol^{-1} dm^{3} s^{-1}; [1 mark]

ii) The missing values are:

The initial rate for experiment 2:

- Rate = 1.19 x 10^{-6}; [1 mark]

$[NO_2]$ for experiment 3:

- $[NO_2]$ = 9.58 x 10^{-2}
  OR
  9.6 x 10^{-2} ; [1 mark]

[Total: 4 marks]

- For part (i)
    - The rate constant calculation only involves $[NO_2]^2$
        - [CO] is zero order reaction, so it is not included in the rate equation and is also omitted when calculating *k*
    - Therefore, $k = \dfrac{\text{rate}}{[NO_2]^2} = \dfrac{3.3 \times 10^{-5}}{(4.1 \times 10^{-2})^2}$ = 1.96 x 10^{-2} mol^{-1} dm^{3} s^{-1}
- For part (ii)

- You have to use the value of *k* that you calculate in part (i)
  - If your value of *k* is incorrect you will lose the mark in part (i) but may score marks in part (ii) by using your value of *k* in the correct calculations (error carried forward)
- The initial rate in experiment 2
  - This is calculated using the rate equation as given with your value of *k*
  - Rate = $k\,[NO_2]^2 = 1.96 \times 10^{-2} \times (7.8 \times 10^{-3})^2 = 1.19 \times 10^{-6}$ mol dm^{-3} s^{-1}
- The concentration of NO_2
  - $[NO_2]^2 = \frac{\text{rate}}{k} = \frac{1.8 \times 10^{-4}}{1.96 \times 10^{-2}} = 0.0091837$
  - When it comes to calculating the missing $[NO_2]$ you need to be aware that this is a second order reaction and you need to find out the concentration of $[NO_2]$ not $[NO_2]^2$
    - You, therefore, need to **square root** the value for $[NO_2]^2$ to find just the value of $[NO_2]$
    - A lot of students forget this final step when a reaction is second order
  - $[NO_2] = \sqrt{0.0091837} = 9.58 \times 10^{-2}$ mol dm^{-3}

5d

d) The rate-determining step is:

Step 1

- (Because,) both reactants are involved in the rate equation; [1 mark]
- (Because,) both reactants have the correct stoichiometry / powers / indices in the rate equation; [1 mark]

[Total: 2 marks]

- If step 2 is identified as the rate-determining step, then no marks can be awarded
- **Remember:** The rate-determining step is the slowest step in a reaction
  - It includes the reactants that have an impact on the reaction rate when their concentrations are changed
  - Any reactants that appear in the rate equation also appear in the rate-determining step and vice-versa
    - In the rate equation, these reactants / chemicals are also raised to the power associated with their stoichiometry for the equation of the rate-